Distributions

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Exercise:

Counting Photons from a Faint Source

A distribution

… tells you the frequency or relative frequency of each possible value or event (or of some data that was collected)

… can be useful for modelling a population of objects

… is a foundation of statistical reasoning

… can be empirical or analytic

… can be continuous or discrete

… can be univariate or multivariate

Analytic distributions have parameters that define its shape

A probability distribution

... characterises random variables

... tells you how probability id distributed across the possible values of a random variable

... integrates to 1

... can be useful of rmodelling a population, your belief in the possible values of a model parameter, etc.

... can be continuous or discrete

... can be univariate or multivariate

... has parameters that define its shape

In practice:

We often need to assume an (analytical) probability distribution in order to be able to analyse our data.

Understanding the properties of the distribution helps us understand what we should see in an experiment.

Random variables

- Random variables are functions that assign a numerical value to the outcome/event from an experiment
- Notation:
- Map from sample space (Ω) to the real line: $X(\omega)$
- Examples of random variables:

Discrete random variables

Continuous random variables

Discrete random variables

- Example: a coin flip

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- We can say $X : \Omega \to \mathbb{R}$ and $X > 0$
- Then the probability for each possible value of X is defined within an interval $P(x \in I)$

Check-in question:

The number of photons that hit a CCD in an hour is an example of a continuous random variable. True or False?

A. True
B. False

Continuous versus discrete probability distributions

Discrete quantities: probability mass function (pmf)

Continuous quantities: probability density function (pdf)

Continuous versus discrete probability distributions

What are some examples of continuous random variables in physics and astronomy?

What are some examples of discrete random variables in astronomy?

Data are realisations of random variables

ADD EXAMPLE HERE!

Population

Sample

Statistics can be used to understand the underlying population, when all you have is a sample

Beware: representativeness matters a lot!

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Sample

- A sample drawn from some distribution
- e.g. the Physics and Astronomy MSc students in Amsterdam, in January
- Will not be exactly like the population because of randomness

Statistics can be used to understand the underlying population, when all you have is a sample

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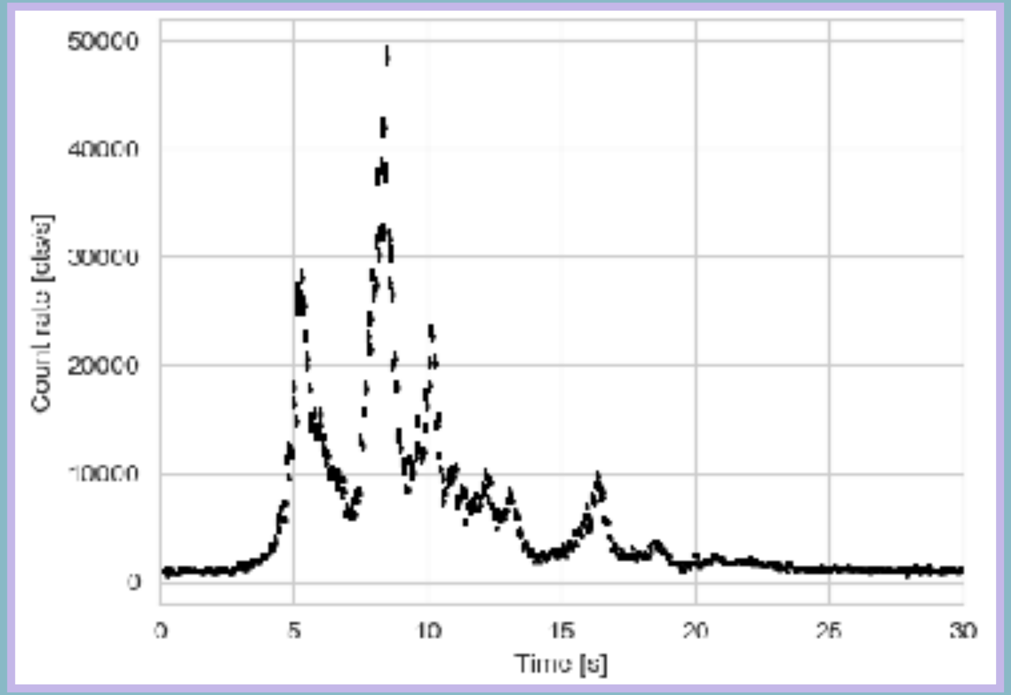
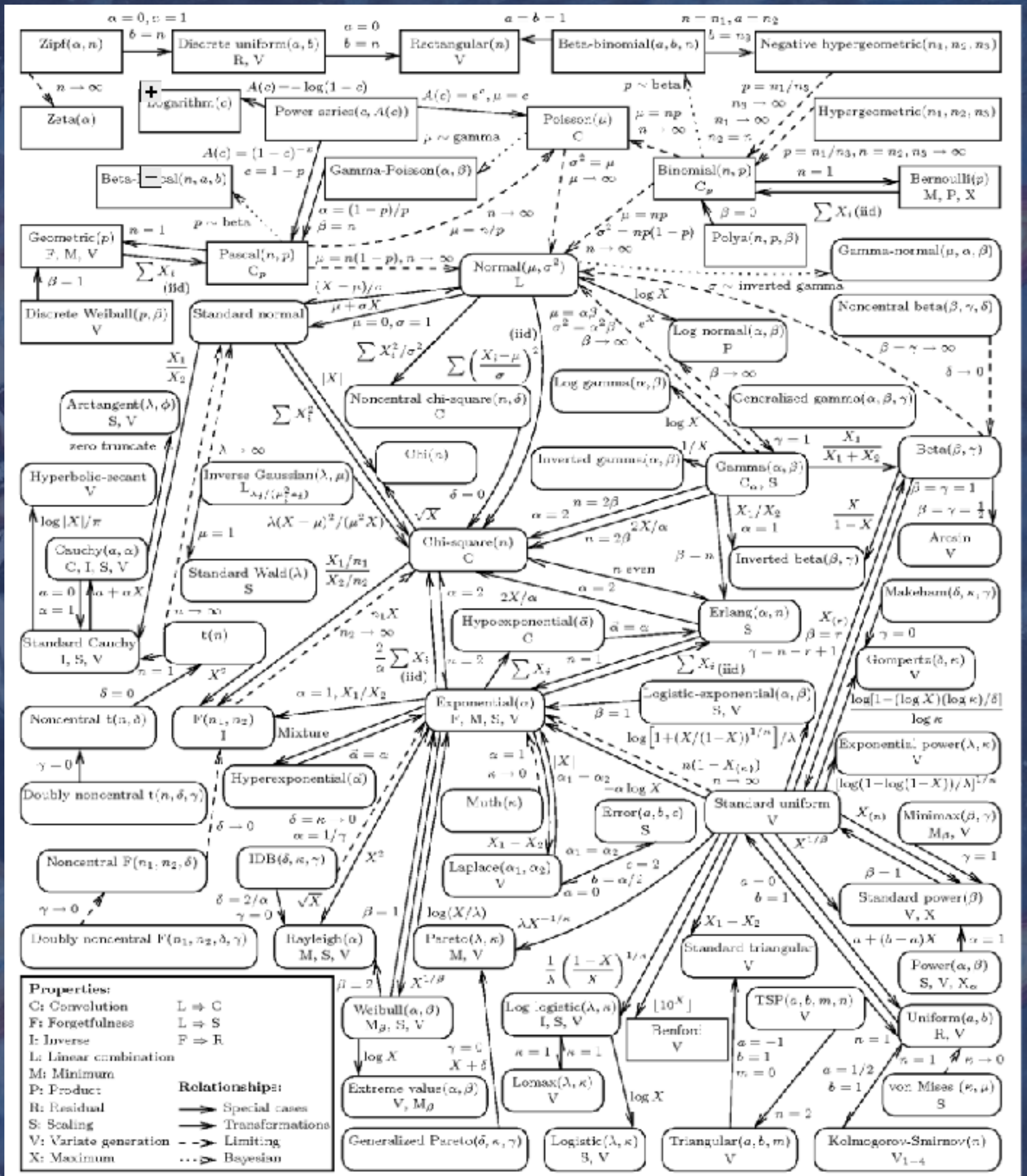
Randomness matters!



- We often don't know how the random variable X is distributed
- Our random data sample x is just that: a sample
- Randomness can trick us! Humans look for patterns in data (pareidolia)
- Things are tricky when data suffer from selection biases, observation bias etc.
- Look at data and summarise it in different ways. Be skeptical!

There are many univariate distributions



https://www.math.wm.edu/~leemis/chart/UDR/UDR.html

What probability distributions do you know?

Common Distributions

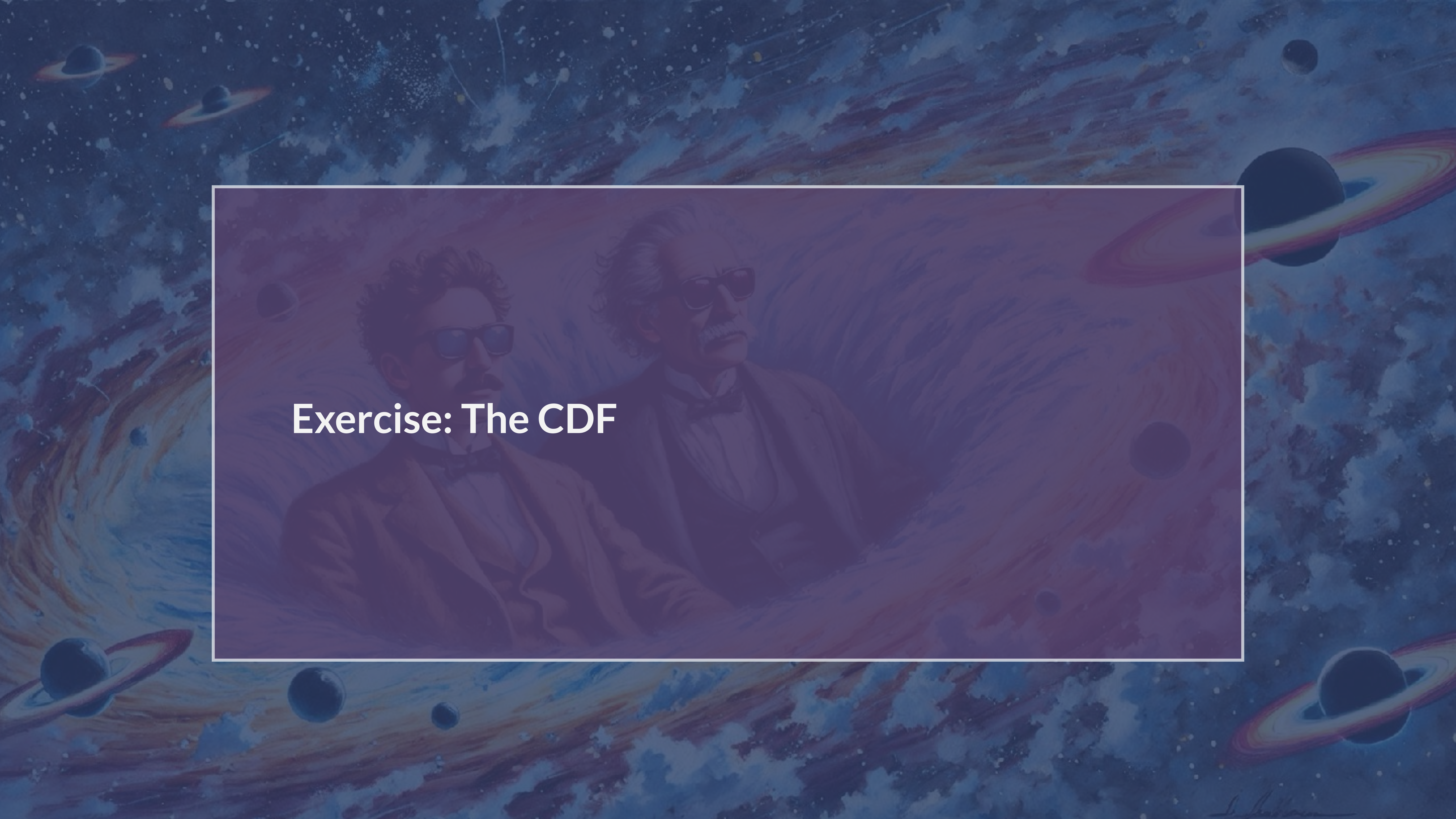
- Bernoulli
  - For binary random variables. Probability of success is p
- Binomial
  - For a sum of Bernoulli n trials, each with probability p
- Poisson
  - For number of events/counts, which happen at some mean rate λ
- Gaussian
  - Naturally arises out of the Central Limit Theorem. Parameters are its mean μ and its variance σ^2
- Student t
  - Used in hypothesis testing for sample means, shape determined by degrees of freedom
- Log-normal
  - When $\log(X)$ is normally distributed, parameters mean μ and its variance σ^2
- Chi-Square
  - When X is the sum of squares of k normally distributed, independent random variables
  - Shape depends on degrees of freedom k

The Normal (Gaussian) Distribution

$$N(\mu, \sigma^2)$$

pdf: $$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{(x-\mu)^2}{2\sigma^2}\right)$$

Exercise: The CDF

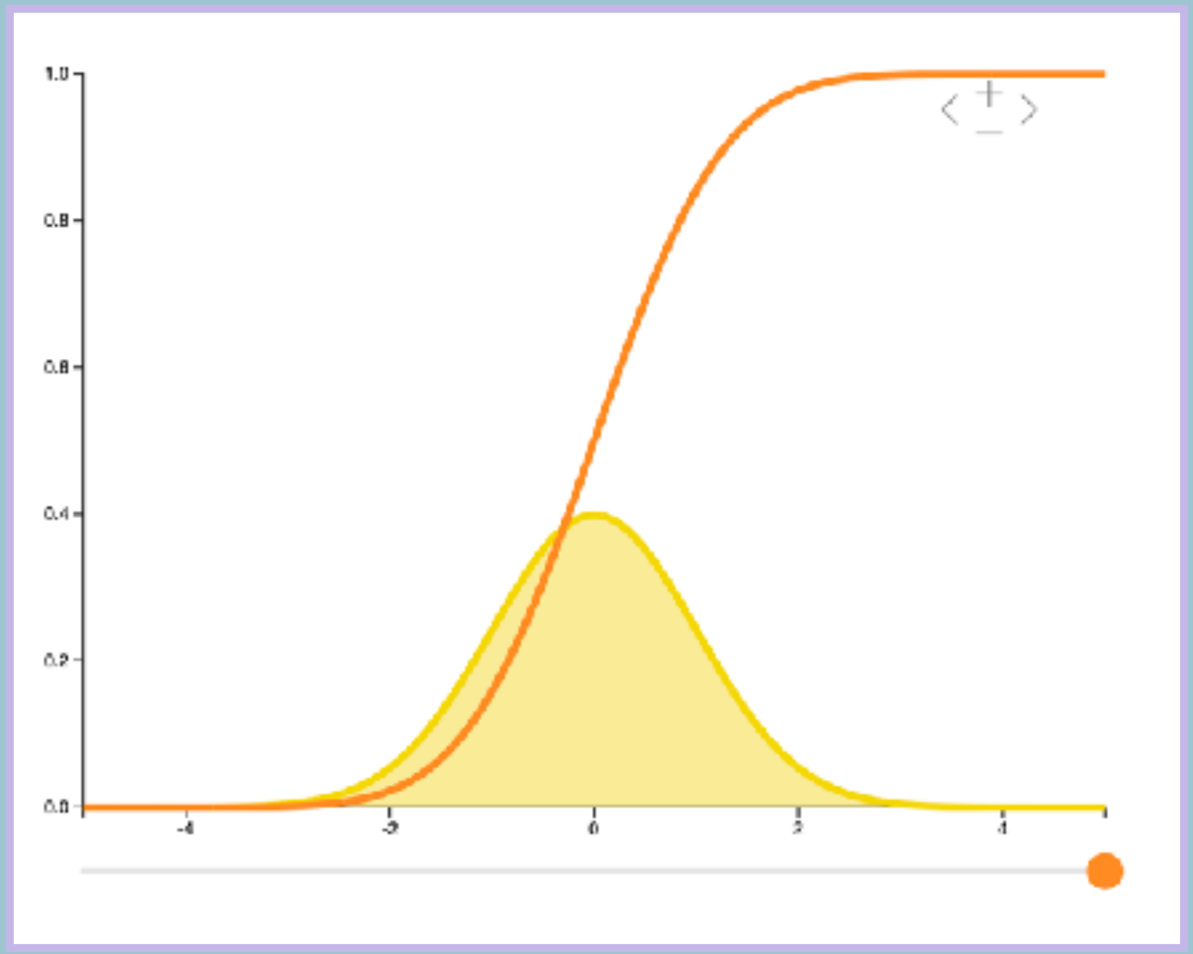


The Cumulative Distribution Function

$$F_X(x) = P(X \leq x)$$

$$P(a < X \leq b) = F_X(b) - F_X(a)$$

$$F_X(x) = \int_{-\infty}^{x} f_X(t)dt$$



https://seeing-theory.brown.edu/probability-distributions/

Percentiles

Quantile: split the distribution into segments of equal probability

Quartiles: 25%, 50%, 75% quantiles

Percentiles: split distribution into 1% steps

- If you're at the 80% percentile on a text, then 80% of students scored below you

- For a sorted dataset $x_1 \leq x_2 \leq x_3 \leq \ldots \leq x_n$, the p-th percentile corresponds to the value at position

$$k = \frac{p}{100}(n+1)$$

- For a probability distribution, the percentile can be calculated using the inverse of the CDF:

$$F_X(x) = p \qquad x = F_X^{-1}(p)$$

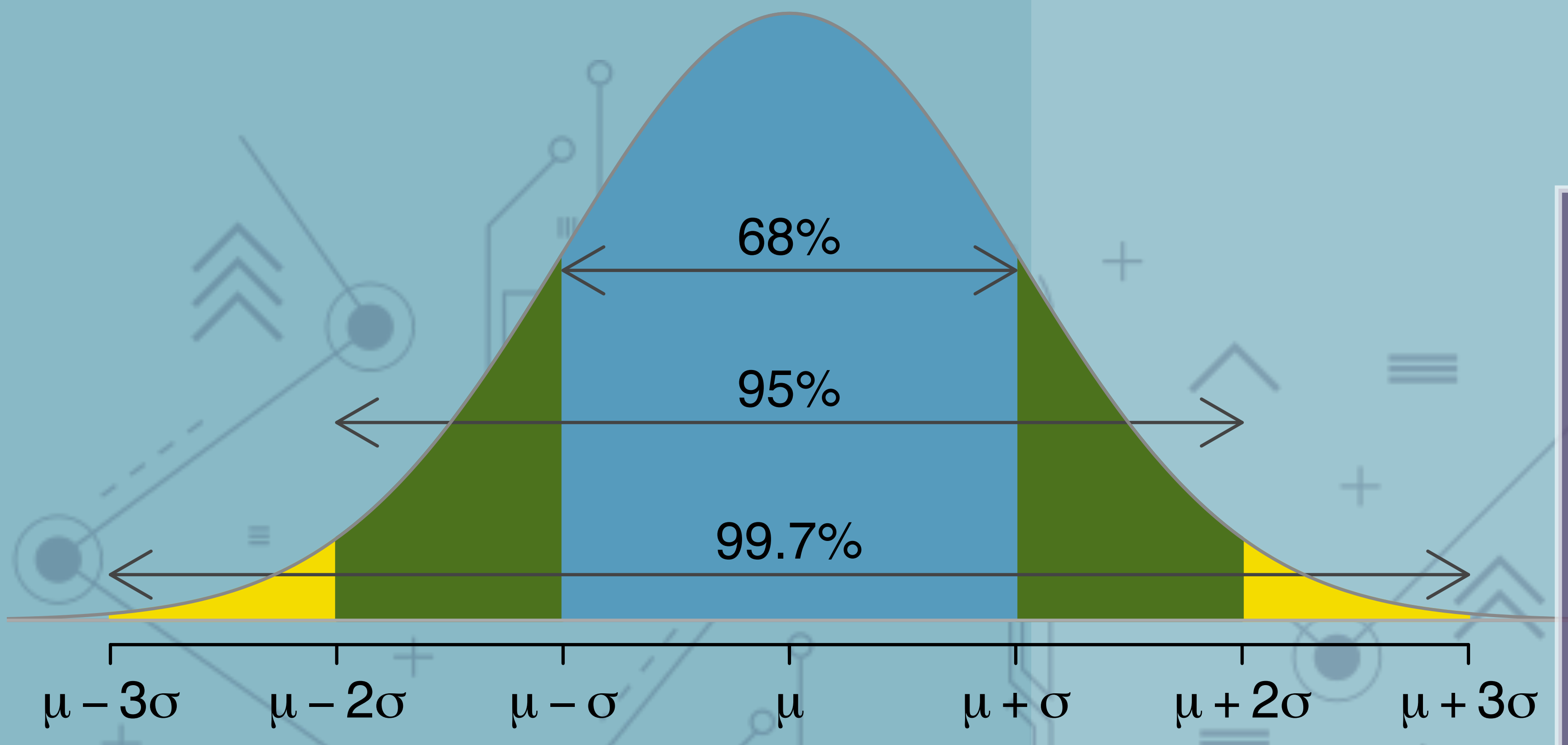
Sigmas

Aka "plot the one sigma errors"

$$N(\mu, \sigma^2)$$

pdf: $$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{(x-\mu)^2}{2\sigma^2}\right)$$

cdf: $$F_X(x) = \frac{1}{2}\left[1 + \operatorname{erf}\left(\frac{x-\mu}{\sigma\sqrt{2}}\right)\right]$$



Important: the 68% percentile is only equivalent to 1σ in the case of a Gaussian distribution!

Expectation and Variance

$$N(\mu, \sigma^2)$$

E[X]: $$f(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(\frac{(x-\mu)^2}{2\sigma^2}\right)$$

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[TO FINISH]

Bivariate and multi-variate distributions

TO FINISH

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Estimator: statistic that can be calculated from a data sample, and approximated the true value of some population pdf parameter. For a population parameter θ, its estimator $\hat{\theta}$ is biased if $E[\hat{\theta}] \neq \theta$

Estimators

A lot of (frequentist) statistics involves calculating estimators, figuring out the statistical distribution they obey, and then comparing the estimators derived from observations with the theoretical distributions they are drawn from.

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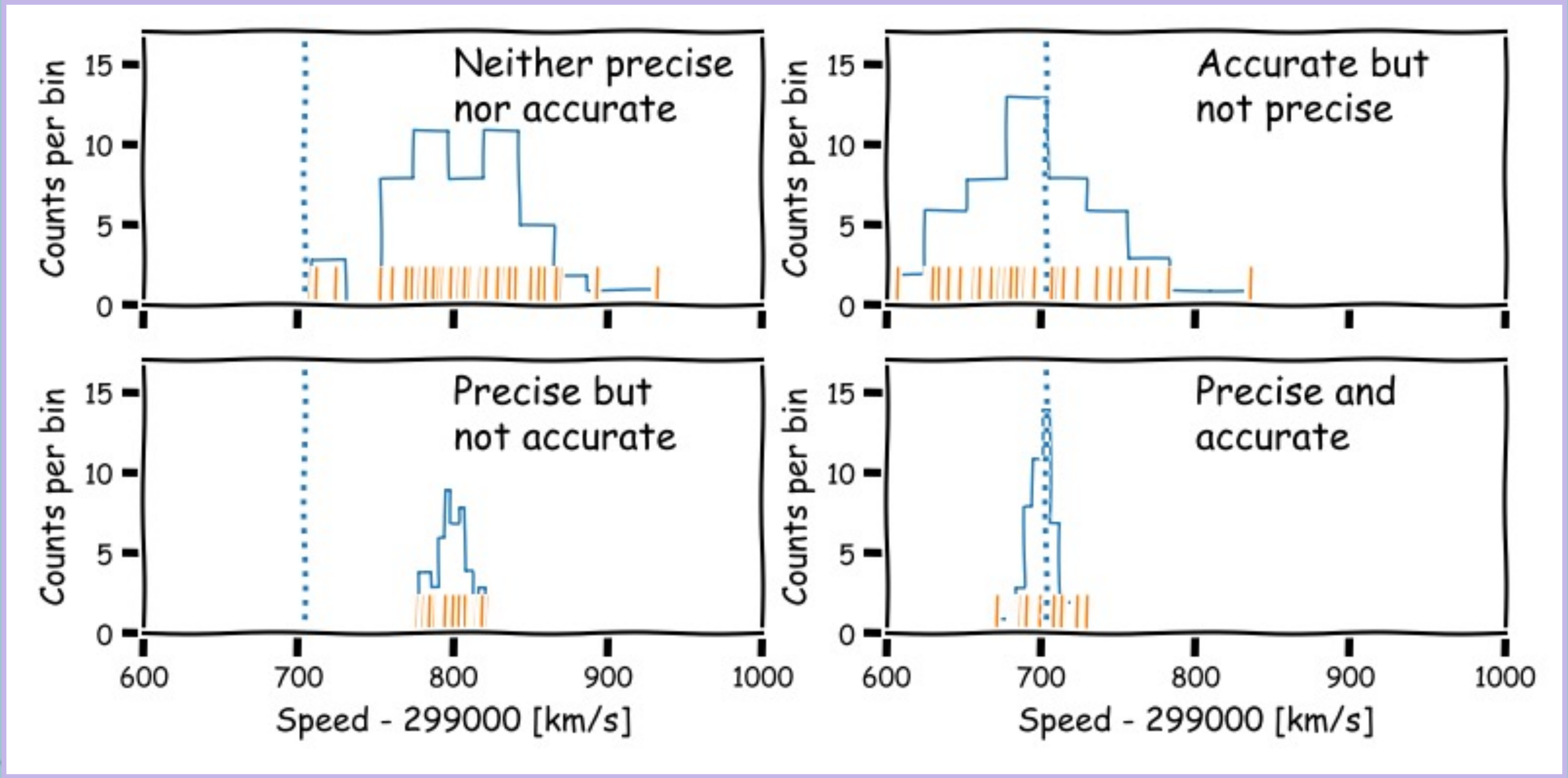
Estimators

Unbiased estimators:

Biased estimators

Accuracy versus precision



Accuracy versus precision

Accuracy relates to bias or systematic error

Precision of measurements relates to noise or statistics error